Dr. Judy Wood and MES: the Space Shuttle Columbia Emails

Emails between Dr. Judy Wood and Mathiew Estepho (MES), September 19 to October 31, 2026, oldest first

These are the emails between **Dr. Judy Wood** and me (**Mathiew Estepho, MES**) about the **Space Shuttle Columbia disaster** and the EM 304 test problem she gave her students at Clemson University on February 6, 2003, five days after Columbia was lost. They are in the order they were sent, from September 19 to October 31, 2026 (times are Pacific time).

38 emails: 27 in full, and 11 short **excerpts** from emails about other things that also mention the shuttle video ([...] marks what was left out). The text is as written; quoted replies and signatures are left out.

The video and notes: mes.fm/space-shuttle-columbia · Dr. Wood's test: mes.fm/em-304-problem-1-judy-wood · These emails as a PDF: mes.fm/djw-columbia-emails-pdf

Dr. Judy Wood Sat, Sep 19, 2026 at 6:41 PM

To: Mathiew Estepho (MES)

How was it?

The first part is about the 9/11 show we had both just been on; the Space Shuttle test starts at "I've attached the Space Shuttle problem."

You did great! Thanks for your help! You were actually amazing, getting up a video or picture as soon as the topic was mentioned.

I hope I didn't bring up too much stuff. There was a variety, so that when we came upon something he seemed interested in, we could focus on that.

Stuff to cover more in the future..

Seismic --- the charts on my website seem good.
Bible
wading through water on West street (no high heat).

I hope I didn't overwhelm him. I was a little worried about bringing up Danny, but it is a category of data. After that, he seemed interested in figuring out what happened to the people.
That video you showed about the the fire chief (?) losing so many of his people who "vaporized" was excellent. It didn't leave me feeling weird about bringing up Danny. And this information really needs to be understood.
Tye made a good point. They probably were dustified quickly and didn't suffer. I hadn't stopped to think about that aspect. It does seem like a better way to go than being trapped somewhere with burns or broken legs and have no one find you. I like Micky Kross' phrase of "existential isolation." Well put.

Anyway, thanks for all your help!

I've attached the Space Shuttle problem. I work out the solution during the test. I also have the "price list" on it, for how many points each part is worth.

Two months after the test, NASA came up with an updated location of the failure. (Now it matched mine. (grin)) Also, they had a thermocouple at the location where the heat ate through the wing box and into the wheel well. Note the temperaturs. Hehehe..... Just what the answer on the test said it would be. :-)
The students who were paying attention really really admired that.
When the final report came out several months later, there was a student doing a work study at an engineering company. He called me that day and said he'd kept his test paper and ran around his work place showing it off. (He had the right answer.)

I think I told you that my nickname was "The Disaster Queen.

I picked the failure location based on where it looked like the foam hit the wing. The fact that the foam made a right-angle turn when it hit told me that it hit hard. You couldn't see exactly where it was because it was on the other side of the wing from the camera (under the wing), but it dustified when it hit. Looks like I picked it correctly. NASA didn't have it correct for the first two months until they found the equivalent of the flight-data-recorder. That thermocouple showed that's where the failure was.

The Shuttle was destroyed on Feb. 1, 2003. This test was given Feb. 6, 2003.
You can see by the detailed drawings that this event really bothered me. They knew the foam damaged the wing. A care package was sent up to the space station Feb. 2, a day later. Why not just keep the shuttle in orbit and re-aim the care package to go to the shuttle. (Care package = oxygen, food, supplies....) And then they could send up another care package for the space station a few days later.

This was the standard cover sheet (except for the bottom).
I didn't start the clock until everybody understood the problem. I assumed it would be scary for them because it was a "real" problem and not a cartoon problem. Part of what was being tested on this test was to draw stresses on a cube, bearing stress, shear stress, and compressive stress, and thermal expansion.

Answer for the lowest failure temperature, 205 deg. F
Note that the failure temperature from the thermocouple reading was about 200-210 deg F on the NASA chart.

:-)

It showed the students that simple calculations can get you pretty close. :-)

G. Scott Hubbard is an experimentalist, like I am. So when the panel met, I watched from his perspective.
I think he had this answer, himself, from the start because of something Harold Gehman said (the leader of the investigation panel).
When they did the final test (shooting foam at a real wing) and it made the exact same damage, Scott Hubbard called up Gehman to tell him the results of the test. Gehman said to Hubbard, "I beg you're

grinning ear to ear right now." That's why I think Hubbard had it figured out early on. But the panel went trough all sorts of fault trees. It was an excellent panel!

Sheila Widnall, whose area of expertise is dynamics. She was fascinated by the 2nd day object that followed the shuttle after it did it's back roll maneuver. Then it started to tumble out of orbit. Based on the way it tumbled, it's orbital decay rate, etc., she figured out what size and shape it was. It was the missing chunk that was punched out of the wing. When they did the roll maneuver, that chunk fell out of the wing.

With Hubbard's final test, shooting foam at a real wing, it punched out the same size chunk of wing and it went into the wing. It would have fallen out with a roll maneuver.

I've only included the shuttle problem. I don't know where the other three problems are. But this is the fun one. :-)

DJ

Attachment: 0301_EM304-T1-1-shuttle.pdf (1,992 KB). Dr. Wood's EM 304 test problem with her worked solution. It is published at mes.fm/em-304-problem-1-judy-wood.

Mathiew Estepho (MES) Sat, Sep 19, 2026 at 10:28 PM

To: Dr. Judy Wood

Re: How was it?

Was a great show! You did great too. Ty seemed like he was holding back a bit but he gave me a lot of leeway to share what I wanted, which was great! I was going to clip out that NYPD Commissioner Bernie Kerik clip before hand, (thanks to Keith for sending us a short clip of that Shawn Ryan interview, so I went to find longer clips). I have just clipped 5 minutes minutes of it, interestingly he also says the dust was sound proof, and also compared it to being in a nuclear blast zone. https://youtu.be/g9edp1aPOUM

And great you found the Space Shuttle problem! I'm getting flashbacks from solids class in university. I think I will cover your test in the next math video I do! I'll mention you were the The Disaster Queen haha

But yeah, 9/11 has so many different aspects, that it could take 100 hours of shows to cover them all.

One reply from Dr. Wood on Sep 19, 2026 at 10:47 PM is left out here: it is about the Bernie Kerik clip and 9/11, not the shuttle.

Dr. Judy Wood Wed, Sep 23, 2026 at 10:29 PM

To: Mathiew Estepho (MES)

Did you do the Space Shuttle Problem Today?

Did you do the Space Shuttle Problem Today? -- on your odcast?

Mathiew Estepho (MES) Wed, Sep 23, 2026 at 11:17 PM

To: Dr. Judy Wood

Re: Did you do the Space Shuttle Problem Today?

Not yet, I plan to start working on it on Sept 30, and it won't be a podcast but a full video + notes as per my usual style. Once a Month I do a long math video, then split it into multiple parts. See notes here of my latest video published August 30th! https://mes.fm/vector-functions-problems-plus (YouTube link: https://youtu.be/YJRehWJxcEY)

Dr. Judy Wood Wed, Sep 23, 2026 at 11:41 PM

To: Mathiew Estepho (MES)

Re: Did you do the Space Shuttle Problem Today?

Have you looked at it yet?

(I shared it with Keith in 2008).

Mathiew Estepho (MES) Thu, Sep 24, 2026 at 12:02 AM

To: Dr. Judy Wood

Re: Did you do the Space Shuttle Problem Today?

yeah I skimmed through it briefly, and got flashbacks from my Statics classes years ago haha I think some if it is more advanced than what I had covered so I'll have to do some research + ask you questions when I get to it. I'll also give an overview of the Space Shuttle disaster. If you haven't seen one of math videos, I try to make them as extensive as possible so they can stand the test of time! (See my Cubic Formula proof, which may be the most (and possibly only) extensive proof on the internet! https://mes.fm/cubic-formula & https://youtu.be/Ne6ITwP6qgo)

Dr. Judy Wood Thu, Sep 24, 2026 at 12:13 AM

To: Mathiew Estepho (MES)

Re: Did you do the Space Shuttle Problem Today?

This was for a Mechanics of Materials class, not Statics.
Statics is the prerequisite.
Some call it Strength of Materials.
This is the fist test of the semester, so it's not much past Statics. But the Statics part is quite simple. It's understanding the formulas for shear stress and normal stress (compressive stress and bearing stress).
The biggest deal is to know what area to put in the formula. ;-)

For this, I showed Roadrummer cartoons in class. That would help them figure out what area mattered.
Like when Roadrunner puts birdseed on a bridge, then cuts a circular hole around the birdseed when the Roadrunner stops to eat. The shear stress area is the circumference of the circle times the thickness. Some idiots just throw in the surface area and guess, hoping for only a few points off. But I told them when I showed the cartoon to remember what surface was holding up that disk of road.. ;-)

The visual of the Roadrummer cartoons really fixed it in their minds. Aother one was the Coyote being an arrow and launching himself. He rams into a cactus. Figure out what area is keeping him from flying through.

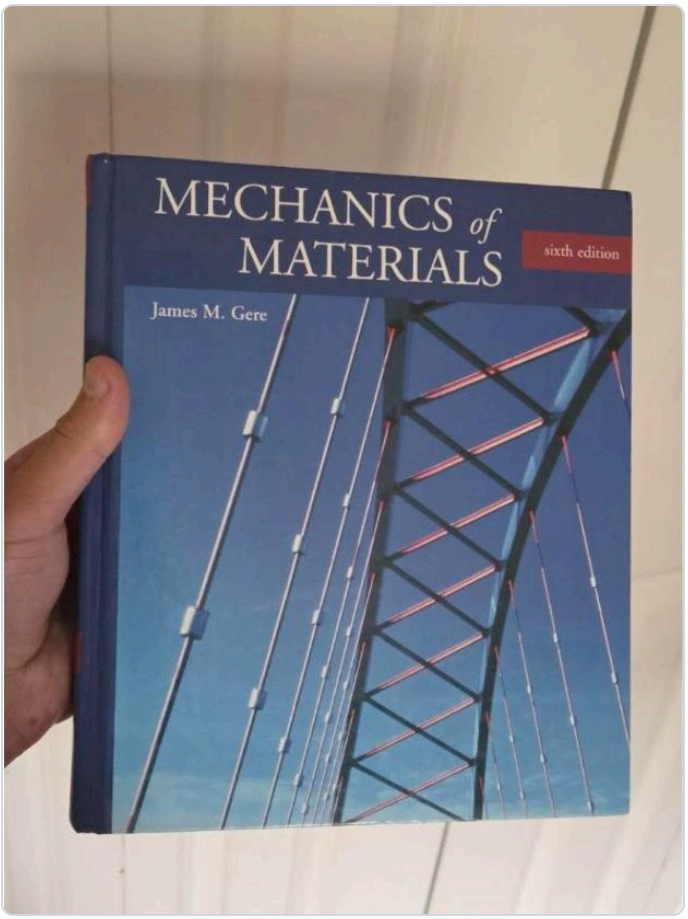
These were examples for shear stress.

Mathiew Estepho (MES) Thu, Sep 24, 2026 at 12:09 PM

To: Dr. Judy Wood

Re: Did you do the Space Shuttle Problem Today?

ahhh yes it's the Mechanics of Mateirals (I called everything related to it as Statics as it was usually static). I still have my MoM book handy! Would've been great to have been in your class to see the Roadrunner physics haha



MES's Mechanics of Materials book (James M. Gere, sixth edition), sent with this email

Dr. Judy Wood Thu, Sep 24, 2026 at 1:47 PM

To: Mathiew Estepho (MES)

Re: Did you do the Space Shuttle Problem Today?

It was funny when I showed a Roadrunner cartoon. It would start playing, and the students were dead quiet. Then they started looking around the room, checking to see if anyone was giggling or smiling and if that was ok. I may have done a little laugh, giving a signal that it was ok to laugh.... then they'd all juoing in laughing.

The one with the bridge, where the Coyote gets a saw and cuts out a circular disk of road aroud where the Roadrunner has stopped to eat bird seed (bait put their by the Coyote)... the disk remains and the bridge falls. Woops, wrong sign convention. ;-)

OH!!!! you may have liked my corny joke in Statics class. It was introducing moments and moment arms. I came in with a big telescoping sea fishing rod. I had a fishing lure on the end, that was a fish filled with buckshot to exactly 4 ounces I'd show him to the class and say this guy here (the silver fish lure) weighs 1/4 pound. So we'll call him "Mr. QuarterPounder." The I start extending the telesoping rod way out until I'd say, "But way out there, he feels like a whopper!"

Corny, I know. But it's something that will stick in their minds.
Before that, and on the second class, we had multiple tug of wars. They were 3-way. On each group were 3-4 roles. A force has a point of contact, a magnitude and a direction. There was a metal ring, and 3 rops extending out. One of the roes had a spring scale. The objective was to calulate the force in the other two ropes. Because this was Statics class, we needed the forces to be constant. So the role of the 4th kid was "Newton Police." He had a big protractor to measure the angle between each rope, requiring equilibrium to measure it (thust Newton Plolice). Either he would write down the angles and the force in one rope, or that was the job of the fifth guy. Then they'd all sit down and figure out how to calculate the other two forces.

I had a few of these activities through the semester. Later on, I had mini teeter-totters, like what used to be at playgrounds. This was desk size. I had a bos of nuts and bolts and otherr heavy objects. They were to place several nuts along one side, then one big one on the other side. Tht is, the sum of the moments on one side = the moment on the other side. There were scales on each side of the mini teeter totter.

Actually, that came to mind when realizing the tipping top of WTC2 appeared to stop tipping. I thought of the little teeter totter with a big weight on one side and a bunch of little nuts distributed along the other side.

When we did trusses, I hd a box of tong depressors, wooden sticks, with a hoe in each end. Make a square, then see how it flops into a parallelogram. At a cross piece, and it beomes rigid. I had severall of these and pased them arond in the class.

Some of them were like I was. It isn't so much what you learn from the thing in your hand, it's about having something to play with in your hand that seemed to make the brain work better. ;-)
So I often brought in "show and tell" items.... just to pass aroud. It made class less boring. ;-) It helps the student feel like he's learning something.

OK, I'll save more stories for later. I bought all the tug-of-war ropes so still have them. ;-)

Your book was not a Beer and Johnston book. Don't knw if you are familiar with those. Anyway, I'm curious what your book's example problems looked like. Weird that this was something I found important.
I had a Mechanics of Materials book, but it looked scary, like a highly technical book written in the 1800s.

Yes, I do think it matters how friendly the book is. :-)

I ended up making workbooks for both classes. They had notes, equations, blank areas to take notes, and example problems. One side was a blank sheet with the FBD on it. The other side had the solution. So you could work out a problem, or sit and watch me work it out while you looked at the solution. (Left side was the un-worked side, right side was the solved side, so both were in view). At the end of each class was a "group work problem." The idea was to help each other start working on a problem and give your new information a test drive. You could get help from each other or me ...whatever you needed. If you just turn the students loose after lecturing, they don't start their homework nutil the night before the next class and they've forgotten stuff by then. So trying out stuff at the end of class helped them get started. For motivatioin, theey were guaranteed to have a

"groupwork problem" on the next test. I would change the numbers and/or needing to solve for something else in the problem. That was bait to motivate them to work on the problem instead of leaving class early.
Golly, I miss those classes.

Yes, I was different from other profs. I loved ti have "hands on" stuff in class. Or visual stuff, like Roadrunner cartoons. It really helps solidify the memory.
At the beginning of Statics class for the semester, I'd say that Statics all comes down to tug of wars and teeter totters. The key is to recognize what you have . (That is, balance of forces, or balance of moments)

My classes were easy -- if you wanted to learn. If you were looking for a free ride, they were hard. That is, you had to learn the material to pass the class. But learning the material was fun and easy.

I did have students who didn't want to be there. Usually girls whose daddy wanted them to be an engineer -- but they didn't want to be one. ;-)

Your book doesn't look beat up. ;-)

I found mine -- and the second edition!!

https://www.amazon.com/Vector-Mechanics-Engineers-Statics-Dynamics/dp/0070042926/

That's a big book, close to 1,000 pages. And knowing you'll have it for both Statics and Dynamics, you know to take care of it. ;-)

I wonder what kind of books they use now? Kindle that you rent? Like my book, you need a hand-held book.

I was trying to find a picture of one of the examples to see the colors.

DJW

Mathiew Estepho (MES) Sun, Oct 11, 2026 at 11:16 PM

To: Dr. Judy Wood
Cc: Keith A L

Re: Interesting excerpt

[...] PS. DJW, I have finally started working on your EM 304 Space Shuttle Columbia test, which I will make an extremely detailed tutorial on it. [...]

Mathiew Estepho (MES) Wed, Oct 14, 2026 at 11:11 PM

To: Dr. Judy Wood

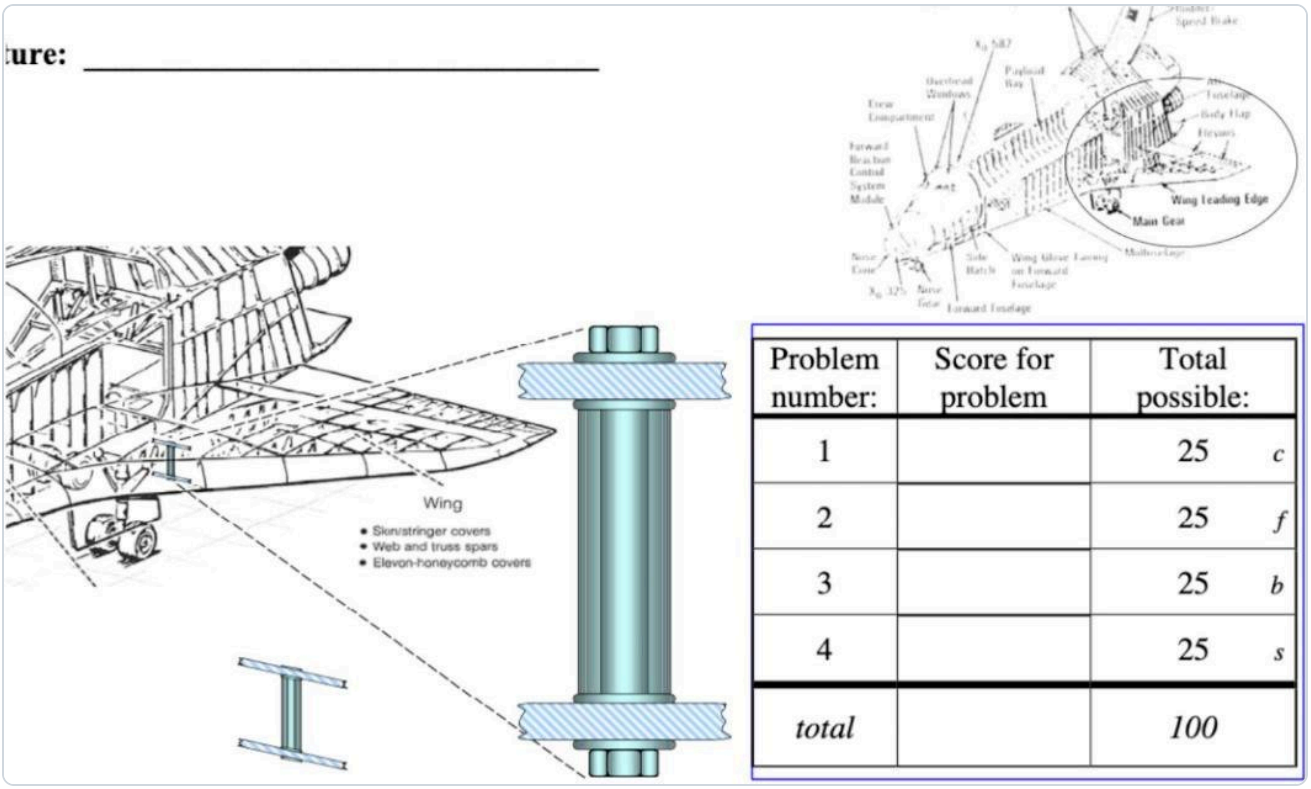
Re: How was it?

Hi again DJW,

I'm going through your Space Shuttle Columbia test and the solution is only for Problem 1. Are there other problems or is this just the test? The PDF says there are four problems on the first page. Let me know if there are other problems + if you can find the solution manual for them too!

PS. I was fortunate to find that Bernie Kerik audio clip, as it helped the audience come to grips with the gruesome reality you were describing...

Cheers,



Problem number:	Score for problem	Total possible:	
1		25	*c*
2		25	*f*
3		25	*b*
4		25	*s*
total		*100*	

The cover sheet of the EM 304 test (with the score table for the four problems)

Mathiew Estepho (MES) Wed, Oct 14, 2026 at 11:33 PM

To: Dr. Judy Wood

Re: How was it?

Ohhh i think problem 1 is just the space shuttle problem, while the other problems are other EM 304 problems etc haha 🤓

Dr. Judy Wood Thu, Oct 15, 2026 at 10:16 AM

To: Mathiew Estepho (MES)

Re: How was it?

I don't know where the other problems are. On my Mac.
This one was saved in a separate file. Also, 2 months later, the CAIB updated their location... to where II and paced it that day. ;-) Those are the back pages. Obviously they weren't with the test.

Dr. Judy Wood Thu, Oct 15, 2026 at 10:17 AM

To: Mathiew Estepho (MES)

Re: How was it?

Yup.

If you need more problems, open a book. ;-)

Dr. Judy Wood Thu, Oct 15, 2026 at 10:30 AM

To: Mathiew Estepho (MES)

Re: How was it?

How is it so far? :-)

My solution is a bit messy. I do the solutions while the students are working on the test. Then, while grating, if I find that students had their solution in a different set up, I'll have that in there, too. You'll see the "price list" for how many points for each part.

That keeps my grading very consistent.
If your work is clear and I can follow it through and determine a mistake was made punching the numbers in the calculator, I might take off 1/10 of a point or less. ;-) (But that isn't advertised.)

I give 4 problems. Some easier and some harder. So I make each one worth 25 points.
That's my way of normalizing the scores. That is, if a hard problem is worth more points, there would be high scores and low scores with not much in between.

You may find a cartoon of a pipe between two walls. I told them if the wing diagram scares them too much to just pretend that it is a pipe between two walls.

I had the correct location of the strike by observing the lift-off video. You don't see the foam hit the wing, but you see that it makes a right-angle turn and dustifies. So obviously it wasn't a light grazing.

Mathiew Estepho (MES) Thu, Oct 15, 2026 at 12:07 PM

To: Dr. Judy Wood

Re: How was it?

Looks good so far. I am currently working on the logistics.... aka using Claude AI to get your PDF on my site as its own page: https://mes.fm/em-304-problem-1-judy-wood

I also made a free online PDF editor, better than Adobe too hahaha 🤯 https://mes.fm/pdf

Also, in the tutorial, I would like to include a write up at the end of how you solved the disaster before NASA did. I will also read that out. And yeah I will have to open my Mechanics of Engineering book to review on some terms haha

PS. I like how you gave the students some diagrams already shown in the empty page. I'm not sure many teachers did that back in my university days haha

Dr. Judy Wood Thu, Oct 15, 2026 at 12:31 PM

To: Mathiew Estepho (MES)

Re: How was it?

It was fun when giving the test. Some of the good students came up with a temperature ruse of 205 F and thought that was a really low number.... and so checked their work a few times. Then they came up to me (during the test) and wondered if they did something wrong because it was such a low number. I smiled and said, "maybe that's what it was."

It's the beauty of simplifying the problem with approximations to get a ballpark idea of what was going on that I was trying to teach them.

Notice in the last two pages are from the April 15 meeting, and were presented by G. Scott Hubbard. Watching the progress of the CAIB panel, I lived through that guy. He's also an experimentalist. I'm guessing he also saw it the way I did. The head of the panel (Gehman) wanted everyone to go through their fault tree, looking at everything. At the very end, when Scott Hubbard set up the final test, first on a fiberglass mock up, the the actual wing of another shuttle that had flown a similar number of f;ogjts. Hubbard set it up perfectly. With the fiberglass wing, he saw he had the aim correct. (You only have one go at the real wing.) Then when he ran the final test and on the real wing, he produced the exact same damage as was on the Shuttle..... as determined by the piece that fell out when the Shuttle did it's roll. Hubbard threw up both arms like an official does to note a "touchdown" in football. When he called Gehman on the phone to tell him the results, Gehman said to Hubbard, "I'll bet you're grinning from ear to ear right now."
For me, that sounded like Hubbard had predicted this all along -- and Gehman knew that.
I'd feel the same way. I think that was an amazing test Hubbard set up.

Dr. Judy Wood Thu, Oct 15, 2026 at 12:43 PM

To: Mathiew Estepho (MES)

Scott Hubbard

I haven't found the video of the test or the real event at the launch.
But here is Scott Hubbard talking about the test.
https://www.c-span.org/program/news-conference/shuttle-columbia-accident-investigation/118700

Dr. Judy Wood Thu, Oct 15, 2026 at 2:30 PM

To: Mathiew Estepho (MES)

Memories

I taught Statics on MWF, and EM304 was T,H, 2 and 3 days after the Shuttle came apart.

Two days and three days after the Shuttle came apart, I walked into my class and said, "Most people my age and older can remember where they were when they learned that JFK was shot." I pointed to the approximate seat where I was in first grade. Instead of using the intercom/speaker in each room, the principal came to our door and talked with our teacher. She burst into tears. I told my 2003 class that I didn't know who this JFK dude was, but something bad awful had to have happened to make our teacher cry like that.
Then I told them that they would remember this day (2003) when their teacher was really upset and angry. I explained that I had wanted to be an astronaut and cold relate to those people. The ground crew KNEW of the impending disaster and DID NOT RELAY THE SERIOUSNESS to the crew, They had the crew download all of their data before the initiation of descent. The crew was seen as expendable.

I think I first said that the cause was the same as the blowing up of the Challenger. That is, politicians making engineering decisions.

When I explained that the foam punched a hole in the wing, destroying the protective tiles, a few students said, "The shuttles lose tiles all the time." I abruptly said, NO, not before or during re-entry. Maybe when they touchdown on the runway on earth, but they must have those tiles through re-entry. A student wondered why. I said because of thermal expansion, if nothing else. Thermal expansion can tear something apart. Consider a can of pop in the freezer. Why does it explode?
Then I said, "Hey, you guys have just learned about thermal expansion. You have a test on Thursday. Maybe I'll put this problem on your test. ;-)

This is how it began.

The Shuttle problem is fairly easy. It's just about knowing what equations to use and what numbers to plug them into.
I really wanted the students to learn how "back of the envelope" calculations (as such approximations are called)) are very useful.

How did they do?
About 1/2 got it 100% correct.
Another 1/3 kinda got it.
And about 1/3 didn't even try. "If NASA can't figure it out, how do you expect us to?" <<== This is the biggest problem.

The ones that kinda got it right, typically hadn't paid close attention to the Roadrunner cartoons. That is, for calculating shear, it's a different surface area than the normal stress.
There were a few who came up and asked me. (I help them recall memories. I don't form new memories.) I asked if they remember the Roadrunner cartoons. Bingo. The student now "got it," ;-)

When I showed them the Roadrunner cartoons, I said that lazy students often jjust throw some area numbers into the equation. They don't know which area is correct, but assume they will get "partial credit." I told them NOPE. After watching these cartoons, you should know what area to use.

No, I was not the typical teacher. ;-) I really wanted the students to learn the material. I also wanted it to be kinda fun.

At Vs. Tech, I taught the equivalent of EM304 testing lab. Clemson didn't have a lab with EM304, but should have.
At Va. Tech, we called it "Busting Lab." We measured the failure strength of materials and how they broke. To add excitement, I'd place a 1-cent or 5-cent bet on the failure load. The winner was the closest without going over. Not required to bet on it, but it made it fun. You know they remembered how things turned out, too.
There were other tricks I'd pull. Cast iron would break with a bang. First test ductile steel.. There was a safety plexiglass around the specimen. So when we were testing Cast Iron and students would get a bit close, when it went BANG! --you can bet every student remembered that. ;-)

Examples I gave for what you'd want a ductile failure or brittle failure with.... Re-bar in bridges..... they start sagging and you know to schedule bridgework the next summer. For brittle failure, I used the example of bolts holding on airplane engines. If the bold is to fail, you want that engine gone,. You don't want it hanging at some weird twisted angle and moving around. That came to mind with the recent disasters of engine dropping off a plane on take-off. But there's no way to deal with that. But if it happened shortly after takeoff, or higher, it would have been ok.

OK, so much for remembering the good times.

Oh, ya.... In Statics class, on the first day, I explain that "everything comes down to tug-of-wars or teeter-totters." I call it a teeter-totter. I think others call it a see-saw. In any case, the kids now (at least in the USA) don't learn what teeter-totters are. They are outlawed. Same with merry-go-rounds and monkey bars. And playgrounds are filled with rubber chunks, not pavement. God forbid that someone would get a skinned knee..... ;-)

Normal young kids bounce. They don't shatter. It's the best time for them to learn about the world.

Did you have merry-go-rounds and/or teeter-totters and/or monkey bars as a kid?
What did you learn from those tings?

DJ

Dr. Judy Wood Thu, Oct 15, 2026 at 3:21 PM

To: Mathiew Estepho (MES), Keith A L

I FOUND IT!!!! The view of the foam on STS 107

I found it!!
The first clip of the foam strike in this video is what I remember seeing on TeeVee. This is the clip I used for my observations. (I think this is the video.) I think it was in Black&White then, though. But maybe my TeeVee was a B&W one. Anyway, based on that clip, I made the test problem for EM304. That is, where the strike hit, and that it was a significant hit.

This is how I selected where the strike occurred and also noticed the right-angle turn and dustification.
The right-angle turn, to me, meant it was not a light grazing. It was a hard hit. Also, after the strike it was dustified.

https://www.youtube.com/watch?v=IgQ3ekcvyRA

I did not see the view from the under-side. The first video was all that I saw on my TeeVee.
But I think it's kind of obvious that the strike was not close in to the "fuselage" of the shuttle, where NASA had said it was. That was weird. The April 15 presentation was after they found the equivalent of the "Flight Data Recorder" for the shuttle. They updated the strike location based on that new data.

Later in that video, they show one of the Scott Hubbard tests. I don't know if this is the real wing or the fiberglass wing. It says it is carbon-carbon, implying it is the real wing, but it's weird it doesn't say that.
(I didn't look close enough.) Also, it doesn't show the set up at a glance.

At 1:20 in the above video:
https://youtu.be/IgQ3ekcvyRA?t=80

In flight, when the shuttle rolled over, something came out of the wing. It wasn't noticed until the second day. It traveled in orbit with the Shuttle but gradually started to fall out of orbit. Based on how it tumbled and its orbit decay rate, Dr. Sheila Widnall (CAIB board member, prof from MIT -- who specializes in dynamics) was able to determine the shape and mass of this object. When they did the test (as shown in the video),, the mass, size and shape of the chunk punched out of the wing was the same as what Dr. Sheila Widnall had determined. It was amazing!!! She's another hero of mine, along with Scott Hubbard. It's weird that it seems such competency is rare these days.

DJ

Mathiew Estepho (MES) Fri, Oct 16, 2026 at 11:07 PM

To: Dr. Judy Wood
Cc: Keith A L

Re: I FOUND IT!!!! The view of the foam on STS 107

Great stuff, thanks for the links! Interestingly, I saw much of these as I was writing up the notes for the video, before reading your email haha Will send you a draft when I'm done the video, hopefully sometime next week if I can remove a bunch of distractions going on recently haha

PS. Yeah it's reinforced carbon-carbon:
https://en.wikipedia.org/wiki/Reinforced_carbon%E2%80%93carbon

Dr. Judy Wood Sat, Oct 17, 2026 at 7:46 AM

To: Mathiew Estepho (MES)

Re: I FOUND IT!!!! The view of the foam on STS 107

That first video, the one I had studied because it was on the TV..... makes you wonder why the NASA folks got the hit location wrong. ;-)

You wrote, "PS. Yeah it's reinforced carbon-carbon:"
The link you gave did not address what I was talking about. I KNOW the shuttle wing was carbon-carbon. Heck I worked on it for my Master's degree! I was referring to the video. I didn't know if that video was from the practice test or the final one with the real wing. The preliminary tests were done with a fiberglass wing segment so ehy could practice everything, first. That video wasn't clear as to which test it was.

Mathiew Estepho (MES) Sat, Oct 17, 2026 at 11:31 AM

To: Dr. Judy Wood

Re: I FOUND IT!!!! The view of the foam on STS 107

Oh right, haha somehow I connected your carbon-carbon comment to that wiki link I was reviewing hahaha I usually get all the info and then put them in my OneNote document that I will then review later for making / filming the tutorial. But looking at the videos and comparing with your exam 5 days later, it's astonishing how accurate you were! (part of me thinks NASA was sidelining a bit to syphon tax-payer money haha).

Here is the Space Shuttle Columbia disaster wiki:
https://en.wikipedia.org/wiki/Space_Shuttle_Columbia_disaster They include a lot of the clips from those videos, including highlighting the part that flew off and hit the wing:



Columbia on the launch pad, from the Wikipedia article: the red circles mark the part of the external tank the foam came from and where it hit the left wing

Dr. Judy Wood Sat, Oct 17, 2026 at 12:20 PM

To: Mathiew Estepho (MES)

Re: I FOUND IT!!!! The view of the foam on STS 107

The fiberglass test was a "dress rehearsal" before the Atlantis wing. No do-overs with the real wing. I vaguely remember they had 3 fiberglass wings to practice on. In my lab, I call it "doing an experiment on the experiment." You go through all the set-up to see what might go wrong.... and how it all works. The last fiberglass test looked just like the actual wing test. I think most of the wing was fiberglass, but the test area was carbon carbon -- which id why it behaved the same.

I knew that day what the likely problem was. The scheduled test was 5 days later, so I had time to make the drawings. ;-)
I was angry about what happened and wanted to be able to say that my EM304 students could even figure it out. ;-)

A few months after the test, I brought a copy of it to a conference, hoping to see a guy I knew from NASA -- who was involved in some of the behind-the-scenes stuff. I noticed his name on some of the

email copies that were posted. When I proudly showed him the test paper, he replied, "At least there was some good that came from this." ;-) (More to that story, but it will need to wait.)

One of the shuttle crew was accomplished in gymnastics. I figured he'd know how to move his body to look at the wing. They had space suits, but no cord. They could have duct-taped a cord to his boots and let him go look. ;-)

But the sickest part was that they did not notify the crew of the seriousness. There was going to be a press conference with the crew, so NASA told them about an "insignificant" foam strike, in case the media asked them about it.
I remember that there was a "care package" (my term) to be launched soon after the Shuttle took off. It was intended for the space station, but they could easily have re-aimed it to where the shuttle was. That would have kept the crew supplied with food, water, oxygen, etc. until it was determined what to do.... at a minimum.

It was also tragic that three different groups had requested spy-satellite imaging of the shuttle wing. Each rrequest was cancelled -- as soon as the higher-ups found out about it. One group almost pulled it off. But the imaging folks contacted NASA to ask where to send the images. "Images? What images??" And it got cancelled.

NASA management didn't want proof that they knew of the damage. But it was obvious they knew. Imagine how expensive all that would have been. So some idiots decided to cross their fingers and hope for the best.

What I did for my Master's degree thesis was actually for the cargo-bay doors. But I was familiar with the whole outer shell of the shuttle. The composite lay-up for the CBDs was such that the coefficient of thermal expansion would be zero -- if the composite was whole. What I was testing was the change in thermal expansion properties if there was a defect in the laminate. I would jokingly describe it as if the guys setting up the laminate wanted to go to the pop machine to get some pop, so reached in their pockets st see if they had enough change.... and lint came out and fell into one of the layers of the laminate. (In reality, they of course work in a clean room. But enexpected things may occur and it may go unnoticed.

So of course I've had a healthy respect for thermal expansion. Most people don't think of it.
Example: pop cans exploding in the freezer. Frozen garden hoses with water in them. Garden hoses under pressure on a hot summer day. (Ryan blew up my hose by leaving it on under pressure in early August. He thought I was just being obsessive when I insisted he turn off the water and relieve pressure after using it.)

DJ

Mathiew Estepho (MES) Sat, Oct 17, 2026 at 10:45 PM

To: Dr. Judy Wood

Re: I FOUND IT!!!! The view of the foam on STS 107

Wow, thanks again for the insider info on the Space Shuttle disaster, I think you know enough for a book haha And let me know what the extra info you have about the NASA insider, when you want to share it!

And of course Ryan blew up the hose hahaha You should've known he was a spook from that day forward haha Interestingly, I always turn off the garden hose while opening the valve, and then wrap it back up so that it drains as much water as it can, because before the pressure builds up and pokes holes in the hose. And yeah just reading through even the mainstream wiki story on the space shuttle, it's amazing how NASA was made aware of the damage to the wing, but they brushed it off as it will be fine haha https://en.wikipedia.org/wiki/Space_Shuttle_Columbia_disaster

And apparently 40% of the shuttle was recovered! 84,000 parts, nearly 100% of them identified.... Meanwhile at the WTC, there's a few placeholder wheel / scrap of something haha

Here is a good video on the Shuttle: https://youtu.be/EW0_Be0Mnlk?t=219

Here is the WTC "plane parts": https://youtu.be/4vnINOtU9QA?t=4597

PS. At the end of my space shuttle video I want to include comments from DJW. Feel free to write something up that you want included, otherwise I will sift through your emails and see what I can put together haha



The recovered Columbia debris laid out on a grid in a hangar



The WTC "FBI plane parts" dumpster



"That's all the 'plane' parts??"

Dr. Judy Wood Sat, Oct 17, 2026 at 10:51 PM

To: Mathiew Estepho (MES)

Re: I FOUND IT!!!! The view of the foam on STS 107

If you want to call me, I'll tell you the insider stuff, but you probably shouldn't put it in your video.

Mathiew Estepho (MES) Sat, Oct 17, 2026 at 11:02 PM

To: Dr. Judy Wood

Re: I FOUND IT!!!! The view of the foam on STS 107

okay I'll call now

Mathiew Estepho (MES) Sat, Oct 17, 2026 at 11:11 PM

To: Dr. Judy Wood

Re: I FOUND IT!!!! The view of the foam on STS 107

Judy Woodruf February 6, 2003 aired same time as your test on CNN:
https://transcripts.cnn.com/show/se/date/2003-02-06/segment/03

Mathiew Estepho (MES) Wed, Oct 21, 2026 at 11:48 AM

To: Dr. Judy Wood
Cc: Keith A L

Re: Run for cover from orman excerpt

[...] PS 2: I have made a bunch of online tools / calculators / simulators on Mechanics of Materials! It would be great if you two can play around with it and let me know any feedback, bugs, suggestions, etc.! (I made it while prepping for the Space Shuttle Columbia tutorial) https://mes.fm/materials

Mathiew Estepho (MES) Mon, Oct 26, 2026 at 9:38 PM

To: Dr. Judy Wood
Cc: Keith A L, Andrew Johnson

Re: Thoughts excerpt

I had a quick scan through (super busy right now, and have started filming the Space Shuttle Columbia disaster video), but it was nice to see he let you share you screen and present a lot of the evidence! [...]

Mathiew Estepho (MES) Mon, Oct 26, 2026 at 10:52 PM

To: Dr. Judy Wood

Re: Thoughts excerpt

[...] Anyways, going to hunker down and finish the space shuttle disaster recording then send you a draft, hopefully tomorrow and will want to publish this week before my spooks stream!

Mathiew Estepho (MES) Tue, Oct 27, 2026 at 6:37 PM

To: Dr. Judy Wood

Re: Geraldo clip? excerpt

It's added to the DJW list now, it was in the Irrefutable channel. I'll organize it all better next time, almost done filming the space shuttle video! [...]

Mathiew Estepho (MES) Tue, Oct 27, 2026 at 10:52 PM

To: Dr. Judy Wood

Re: Geraldo clip? excerpt

[...] Anyways, I finally finished the Space Shuttle recording! Will do a bunch of editing et al later this week. I also found the Wile E Coyote clip you mentioned about the area for the shear stress haha where he cuts the a hole in the bridge but the bridge falls, gold! https://youtu.be/OmVCFonmslU?t=144 [...]

Dr. Judy Wood Tue, Oct 27, 2026 at 11:33 PM

To: Mathiew Estepho (MES)

Re: Geraldo clip? excerpt

[...] So cool that you found the Roadrunner cartoon with the wrong sign convention. ;-)
There is also one with an ACME steel plate, where the Coyote makes a coyote-shaped cut out.
And there is one with a cactus. Wile E. Coyote gets a big bow and puts himself in it like an arrow and launches himself. He goes flying and then starts to fly through a cactus, but gets stuck. So, calculate what surface area stopped him. Wile E.'s circumference times the thickness of the cactus (as an approximation).
Golly those were fun videos.
I think you can see how they leave a picture in the students mind they won't ever forget.
;-) [...]

Glad you got your video done.

You're good at the graphics! ...or telling the AI the right stuff to get it to turn out nicely.

Mathiew Estepho (MES) Wed, Oct 28, 2026 at 8:33 PM

To: Dr. Judy Wood

Re: Geraldo clip? excerpt

[...] Yeah that roadrunner cartoon was epic. I used to watch it a lot when I was younger, but I had not seen that circular hole in bridge clip before haha

I have just uploaded an Unlisted Draft of the video (I need to go over and make timestamps / etc. before publishing). It's pretty extensive so let me know if you watch it!
https://youtu.be/x2kf5eITPRo

The graphics I made myself for the thumbnail (besides the Undercover clown AI haha), and I used my free online photo editer: https://mes.fm/photo Play around with it and let me know any bugs / issues / features you need improved! [...]

Mathiew Estepho (MES) Thu, Oct 29, 2026 at 11:33 PM

To: Dr. Judy Wood

Re: Hello? excerpt

[...] PS. I'm working OT to publish the Space Shuttle Columbia video. it's done, just working on the notes. Here's the final version video (will publish when I get the notes up)! https://youtu.be/zNdifMfnqcs

Mathiew Estepho (MES) Fri, Oct 30, 2026 at 12:05 AM

To: Dr. Judy Wood

Re: favor? excerpt

[...] I have just published the space shuttle tutorial, and have published the notes too! https://mes.fm/space-shuttle-columbia

I think this video will showcase to people the extensive knoweldge The Disaster Queen!! [...]

Mathiew Estepho (MES) Sat, Oct 31, 2026 at 11:57 AM

To: Dr. Judy Wood

Re: Steven Jones responded to you excerpt

[...] Interestingly, when watching news footage of the Space Shuttle Disaster, a person's whole house shook way down below on Earth! https://youtu.be/hkrf8Jjv0tw?t=120

Published at mes.fm/djw-columbia-emails